
Neural Operators for Disruption Prediction in Nuclear Fusion Plasmas

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Abstract

Nuclear fusion has the ability to combat climate change and decarbonize the power grid, but the detrimental effects plasma disruptions have on multi-billion dollar fusion tokamaks hinder their commercial viability. Although modeling disruptions through physical simulations is very challenging, machine learning (ML) models have achieved remarkably high accuracies in predicting these phenomena. In this work, we introduce Fourier Neural Operators (FNOs) for disruption prediction, achieving a true positive rate of 96.2%, which gives us good insight for the next step of future disruption prediction, for which we aim to surpass the critical 95% threshold required for commercializing tokamaks. Our approach leverages time series data from three different tokamaks to predict disruptions, and takes advantage of the discretization invariance of FNOs to generalize across machines with varying sampling rates. We demonstrate that the FNO outperforms baseline models (CCNN and Transformer) when trained and tested on the C-MOD tokamak, achieving an AUC of 0.980. We also explore the importance of various features to enhance model interpretability. We identify a trade-off between sampling rate and memory efficiency, showing that downsampling reduces memory load but can cause the true positive rate to reduce. Finally, we analyze how the length of input sequences impacts performance, identifying the optimal temporal window needed for reliable forecasts.

1. Introduction

Nuclear fusion power plants exhibit considerable potential to combat climate change due to their ability to produce carbon-free energy without generating high-level radioactive waste or requiring substantial land use while still producing immense amounts of energy (Giuliani et al., 2023). This work considers tokamaks, which use magnetic confinement to generate a large current through the plasma and aim to produce net positive energy as shown in Fig 2 (Spangher et al., 2023). These reactors have the highest level of techno-

logical readiness to produce controlled nuclear fusion power (Wurzel & Hsu, 2022). However, despite their symmetric shape, magnetohydrodynamic (MHD) instabilities can cause plasma to exert unequal amounts of magnetic force on the field, which eventually causes the plasma to escape confinement, known as a disruption. Disruptions are instabilities that terminate the fusion reaction and cause severe damage to the walls of the tokamak due to the dissipation of the extremely high temperature and current in the walls (Spangher et al., 2023). With the risk of disruption, tokamaks are not commercially viable (Wurzel & Hsu, 2022). Modeling disruptions using physical simulations is challenging, as it is computationally complex to model macroscopic and microscopic phenomena simultaneously and to encompass the many known and unknown complex factors that influence plasma dynamics (Spangher et al., 2023). Hence, at the moment, we rely on data-driven learning.

ML models have been proposed as a method to forecast the likelihood of plasma disruptions. However, a true positive rate above 95% is necessary for commercial-scale tokamaks (Maris et al., 2023), which, to the best of our knowledge, has not yet been achieved. Ultimately, disruption prediction models are designed to be integrated into a larger control system that activates preventive measures when an impending disruption is detected to mitigate the damage caused by disruptions.

This work focuses on developing neural operator models trained on time-series data collected from three major tokamaks. We evaluate several data preprocessing techniques, including data augmentation and undersampling. For augmentation, we extract fixed-length windows from the input embeddings and label them as disruptive or non-disruptive based on whether the end of the window overlaps with the disruption window—an interval defined for each tokamak. We also conduct feature ablation studies to assess how the shuffling of specific input features affects model performance. This helps identify which features are most critical to the model’s predictions, improving interpretability and potentially offering insights into plasma behavior. In addition, we investigate curriculum learning strategies to further enhance model accuracy.

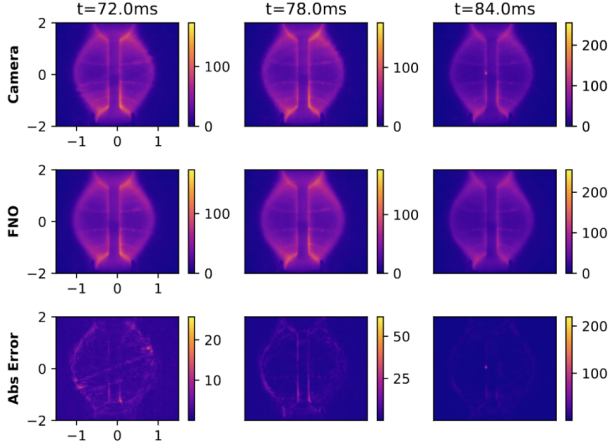


Figure 1. Extracted from (Gopakumar et al., 2024), Comparison of plasma evolution across one feed-forward operation of the FNO spanning 10 frames ~ 12 ms. top: actual data from the experiment, middle: emulation performed by FNO, bottom: absolute error across the prediction and the camera. .

2. Related Work

(Gopakumar et al., 2024) demonstrates that Fourier Neural Operators (FNOs) can successfully predict the evolution of plasma, as shown in Fig 1. However, (Gopakumar et al., 2024) does not explicitly seek to predict disruptions, only to perform plasma state forecasting. Further, the data in (Gopakumar et al., 2024) consists of images obtained by cameras positioned within the MAST Tokamak, whereas the (open-source) data (data, a) this work uses is time series data with features such as plasma current, electron density, plasma pressure, etc (see Table 1 for complete list). Moreover, (Maris et al., 2023) suggests that multi-modal FNOs can be applied to camera data from tokamaks to predict variants of the plasma state. This provides a strong precedent for training neural operators to gain more insights into disruption prediction in nuclear fusion plasmas.

Various ML models have been proposed to predict the likelihood of plasma disruptions, including deep learning models (Kates-Harbeck et al., 2019), deep reinforcement learning (Seo et al., 2024), manifold learning techniques (Cannas et al., 2013), autoregressive transformers (Spangher et al., 2023), which leverage long-term memory) (Arnold et al., 2023). Additionally, hybrid architectures such as the Hybrid Deep Learner, which utilizes a convolutional LSTM network (Zhu et al., 2020), have been explored.

However, none of these methods have achieved a true positive rate of at least 95%, which remains a critical benchmark for commercial-scale tokamaks. (Spangher et al., 2023) suggests that this is due to several factors: insufficient fine-tuning to the unique requirements of the physical

problem, an inability to effectively capture the temporal evolution of plasma density and temperature (often referred to as “memory”) (Zohm, 1996), and the reliance on data sampled at fixed rates, which constraints the models from detecting subtle variations in the data. In contrast, CCNNs are a type of neural operator that learn interpretable, continuous filters with longer ranges than those achievable by discrete models (Arnold et al., 2023) and have achieved good performance for this task. However, the training samples used were extremely short (less than a second). We explore the FNO architectures for disruption prediction, alongside various data preprocessing techniques, data subsampling, and initialization methods to enhance model performance.

3. Methods

3.1. Modeling Task

3.2. Model Architecture

CCNN was a first attempt to take into account the variability of the data format (i.e., different sampling frequencies across tokamaks) (Arnold et al., 2023). We extend this by performing a more thorough investigation of the use of neural operator architectures for this sequence to label (disruption/no disruption) task, starting with the FNO. For a thorough overview of our disruption prediction framework, please see 2

Since the data’s dimensions are continuous variables, they are treated as embeddings and normalized using the mean and variance of the training dataset. We fine-tune the model’s hyperparameters — including the number of layers, number of modes, dropout rate, hidden channels, and learning rate scheduler — to optimize performance with respect to the cross-entropy loss. We use Bayesian hyperoptimization strategies, such as the Tree Parzen Estimator to select the set of hyper-parameters to be tried after each trial, by taking into account the history of experiments (Wu et al., 2019). All training was done using the Resnick High-Performance Computing Center (HPC)¹ at the California Institute of Technology with GPUs and testing on the Area Under the Curve (AUC) as this is the main metric used for measuring disruption prediction accuracy in previous works. We further explain the main model we explore below.

Fourier Neural Operator (FNO): We use an FNO architecture as proposed in (Li et al., 2020) as FNOs are considered a good starting point for operator learning problems. They can model turbulent flows with zero-shot super-resolution and achieve superior accuracy compared to learning-based solvers under fixed resolution for this task. In essence, FNOs are a class of NOs that parameterize the integral kernel directly in Fourier space, enabling a highly efficient and

¹<https://www.hpc.caltech.edu/resources>

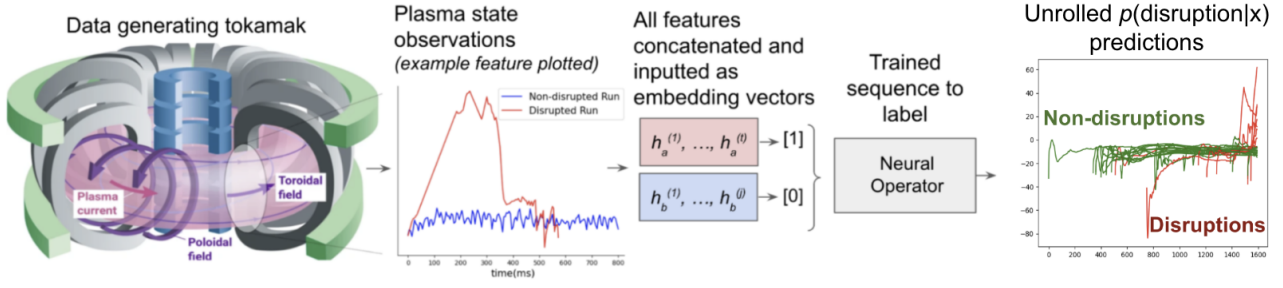


Figure 2. Disruption prediction framework for the sequence to label classification task using neural operators, adapted from (Spangher et al., 2023). Time-series data is generated from three different tokamaks. Note that observation frequencies are only consistent among each individual tokamak. The data is clipped at the 1st and 99th percentiles, then normalized. The model inputs are each a vector of data corresponding to a single timestep, and hence form a sequence of timesteps. These vectors contain raw data values produced by the tokamak for each input feature. Inputs with fixed sampling rates (produced using forward fill) and variable sampling rates are explored, as neural operators are discretization invariant and hence can handle data with varying resolutions. The model outputs raw logits with an additional temporal dimension, which is averaged, and passed through a softmax function to compute the probability of a disruption for a given input sequence. Then, a threshold can be applied to produce a final prediction.

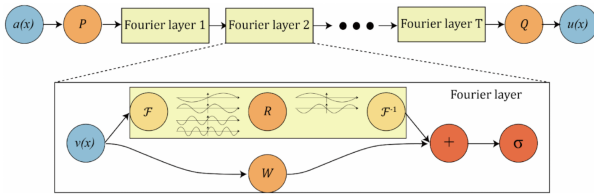


Figure 3. The FNO architecture (extracted from (Li et al., 2020))

expressive architecture. Hence, the FNO model is capable of real-time inference. We leverage the discretization invariance of FNOs to train on shots of varying lengths—due to different sampling rates—over a fixed time interval. This avoids interpolation errors from upsampling low-resolution shots (e.g., via forward fill in (Spangher et al., 2023)) and preserves important high-frequency information in high-resolution shots by eliminating the need for downsampling.

4. Experiments

4.1. Origin and Preparation of Dataset

The open-source dataset, available at (data, a), contains trials (known as shots) collected from multiple tokamaks². This diversity makes the dataset particularly valuable, as it enables investigation into whether a model trained on data from certain reactors can generalize in a zero-shot setting to a different, unseen reactor. Tokamaks are run for a “shot”

²MIT’s Alcator C-Mod, General Atomics’ DIII-D, and the Chinese Academy of Sciences’ EAST tokamak, as described in (Spangher et al., 2023)

which lasts anywhere between seconds to minutes, and this generates high volumes of multimodal data of which we focus on certain global state variables, listed in Table 1. This gives us a time series of those d features, as illustrated in Fig 4, where each point in time corresponds to a vector in $\mathbb{R}^d$. The code used to generate all experiments and models is publicly available³.

It is important to note that each tokamak’s diagnostic equipment records metrics from each shot at a different sampling rate. Therefore, we explore data subsampling techniques, such as normalizing the data across different tokamaks by discretizing at 5 ms and interpolating via forward-fill, as well as training models with data of different resolutions, skipping the discretization step. When we train with varying resolutions, we ensure the data spans a fixed time interval, as the FNO requires inputs of consistent temporal dimensions to apply the Fourier transform and operate in the frequency domain appropriately. Since the control system takes at least 40 ms to take corrective measures, we truncate a minimum of the last 40 ms as predictions made after this would be too late. We also try truncating extra timesteps from the end of each sequence, as (Spangher et al., 2023) notes that these final steps can introduce noise that does not generalize well across different disruptive events. Additionally, we explore data subsampling strategies to determine whether higher sampling rates can be used while still maintaining sufficient predictive accuracy. This approach aims to optimize memory and computational efficiency by using only the necessary amount of data during training.

³<https://github.com/SreeyuR/cs-165-final-project>

Table 1. Input features of the model (Spangher et al., 2023).

Definition	Feature	Relevant Instab.
Locked mode indicator	Locked mode mag. field normalized to toroidal field	MHD
Rotation mode indicator	Std. dev. of Mirnov array normalized by toroidal field	MHD
β_p	Plasma pressure normalized by poloidal magnetic field	MHD
l_i	Normalized plasma internal inductance	MHD
q_{95}	Safety factor at 95th flux surface	MHD
n/n_G	Electron density normalized by Greenwald density	Dens. limit
Δz_{center}	Vertical position error of plasma current centroid	Vert. Stab.
Δz_{lower}	Gap between plasma and lower divertor	Shaping
κ	Plasma elongation	Shaping
P_{rad}/P_{input}	Radiated power normalized by input power	Impurities
$I_{p,error}/I_{p,prog}$	Plasma current error normalized by programmed current	Impurities
V_{loop}	Toroidal “loop” voltage	Impurities

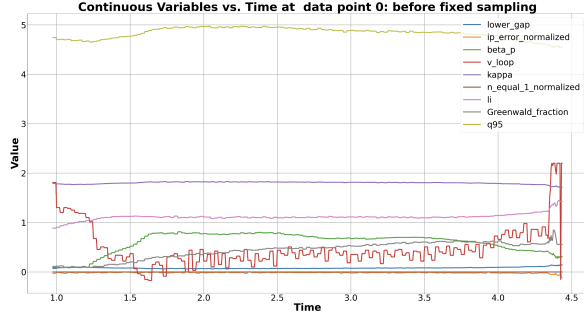


Figure 4. Continuous variables over time for data point 0, with values clipped at the 1st and 99th percentiles of the raw data.

5. Results and Discussion

We trained FNO models and evaluated their real-time disruption inference performance, comparing their AUC against CCNN and Transformer models, previous SOTA models for this task. We trained our models on time series data collected from multiple tokamaks, as described in Section 4.1, and evaluated their ability to generalize to a new reactor with relatively limited data. Following the setup in (Spangher et al., 2023), we designated C-Mod as the held-out reactor for evaluation, while using DIII-D and EAST for training due to their larger number of available shots. This reflects a realistic use case in which data from well-instrumented, mature reactors is used to improve predictive performance on newer or less-instrumented machines. To ensure consistency, the training, validation, and test sets are constructed to have approximately the same proportion of disruptive and non-disruptive shots, attempting to preserve the underlying class distribution. In addition to the cross-reactor generalization setup, we also evaluated model performance in two other settings: (1) training and testing on data from a single tokamak, and (2) training and testing on the combined data from all reactors. The results of these experiments are presented below and compared against relevant baselines.

Our results reveal several key trends. First, the FNO achieves the highest overall performance when both training and testing are done on C-MOD data, with a mean AUC of 0.980, outperforming the CCNN (0.873) and Transformer

(0.910) baselines. However, when the FNO is trained on data from other machines (DIII-D and EAST) and tested on C-MOD, its performance drops substantially, with AUCs ranging from 0.623 (fixed rate, all machines) to 0.891 (fixed rate, DIII-D and EAST only). This suggests a strong degree of machine-specific signal in the disruption dynamics, consistent with prior findings in (Zhu et al., 2020).

Additionally, we observe that the FNO consistently benefits from variable sampling rates, achieving higher AUCs compared to the fixed-rate setting when trained on multi-machine data (e.g., 0.871 vs. 0.623 when trained on all machines, tested on C-MOD). This supports the hypothesis that the FNO is able to leverage the richer temporal resolution and diversity provided by variable sampling rates.

In contrast, the Transformer baseline is limited to fixed sampling rate inputs and exhibits poor zero-shot generalization performance when trained on DIII-D and EAST and tested on C-MOD (AUC of 0.645), falling short of both FNO and CCNN. The CCNN shows more stable performance across settings, but does not match the FNO’s best-case results.

Overall, these findings highlight the FNO’s strength in handling high-resolution, machine-specific data and its ability to generalize better when exposed to varied sampling patterns. This makes the FNO a strong candidate for deployment in scenarios where data from a specific reactor is limited.

Figure 6 highlights the importance of capturing sufficient plasma history for effective disruption prediction. This aligns with the notion of plasma exhibiting long-term memory—having enough temporal context improves model performance, but including too much non-disruptive data can dilute predictive signals, leading to missed disruptions.

6. Conclusion and Future Work

Since we assess the overall accuracy of the FNO and baseline models by mean AUC, since the true positive rate of the FNO model when trained on data exclusively from cmod is 96.2%, indicating it is a very good classifier, with a false positive rate of 2.2%. The next steps would involve training this model to predict autoregressively, so it can predict disruptions well in advance. Then, we can use this disruption model within some other form of control, such as an RL model that is activated to mitigate damage when the model predicts an upcoming disruption. For example, this disruption prediction algorithm could be incorporated into a disruption mitigation system such as ITER.

To extend upon this work, we can use this model to understand the importance of various features by performing ablation studies, as well as exploring this new model architecture to help inform the plasma science community about the nature of plasma disruptions. Furthermore, the use of

Train	Test	Sampling Rate	FNO	CCNN	Transformer
d3d, east, cmod	cmod	fixed	0.623	0.604	0.830
d3d, east, cmod	cmod	variable	0.871	0.636	n/a
d3d, east	cmod	fixed	0.891	0.836	0.645
d3d, east	cmod	variable	0.887	0.747	n/a
cmod	cmod	fixed	0.980	0.873	0.910
d3d	d3d	fixed	0.849	0.874	0.857
east	east	fixed	0.869	0.902	0.845

Table 2. We compare the mean AUCs achieved by Transformer (Spangher et al., 2023), CCNN (Arnold et al., 2023), and FNO-based architectures across various train-test setups and sampling rates for the sequence-to-label task. Unlike (Spangher et al., 2023), we do not use data augmentation, curriculum learning, or state pretraining, as our approach (detailed in Section 4) was sufficient to achieve the desired prediction accuracy. Additionally, since we do not pad sequences to a fixed length—only ensuring a consistent time interval—we cannot evaluate the Transformer on varying-resolution sequences, as that model requires inputs of uniform length (typically achieved via padding). Therefore, we omit the Transformer from those specific experiments.

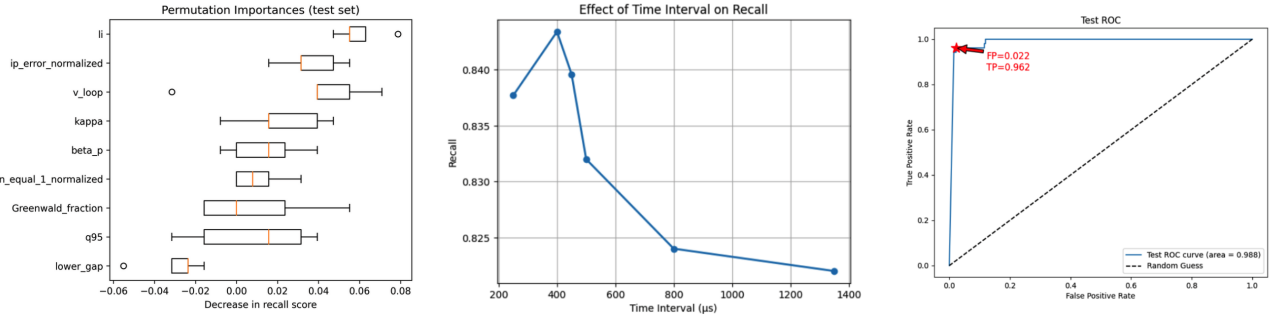


Figure 5. We analyze the importance of different features in our model, trained on C-Mod data, to see if reducing the features can help improve accuracy. The x axis shows the drop in the recall when one feature is randomly shuffled and the y axis shows the continuous features of our model. We can see that the model does not rely very much on some features like li (internal inductance) as there is not too much of a decrease in the score when this feature is shuffled.

Figure 6. Effect of shot duration on disruption prediction. The recall increases as the shot time interval grows, peaking at 400 microseconds. Beyond this point, recall begins to decline, indicating that longer input sequences may include too much non-disruptive information, making it harder for the model to identify disruptions. Ablation study performed on FNO model trained on all three tokamaks, evaluated on C-MOD

Figure 7. The Test ROC Curve. At a false positive rate of 2.2%, the model achieves a recall of 96.2%, meaning it correctly predicts 96.2% of disruptions. This reflects strong performance at a low false alarm rate. The model was trained on data from C-MOD.

NOs for fusion is another real-world application of NOs that illustrates their utility in real data engineering problems.

In order to interpret how well the model’s predictions align with known physics, future steps could also involve correlating disruptivity to confinement regime predictors. For instance, Alessandro Pau’s confinement prediction model could potentially be used to create a confinement label for the existing data. Then, one could correlate the confinement prediction with the disruptivity prediction. If a confinement regime change ($H \rightarrow L$ back transition) correlates with the early disruptivity warning, then this supports the claim that the models are picking up on physics and could support the use of disruptivity models in disruption avoidance. One could also use the confinement label as another feature to train the model with to improve prediction accuracy.

Furthermore, one could try to predict plasma instabilities that do not necessarily end the shot using the disruptivity models. This would be a more generalizable problem and reuses similar architecture as the disruption prediction framework. However, this extension would involve obtaining a training label for instability and modifying the existing dataset with this information.

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